HMCNet: Hybrid Efficient Remote Sensing Images Change Detection Network Based on Cross-Axis Attention MLP and CNN

Liejun Wang and Haojin Li

***Abstract*— As an important task in the field of remote sensing image interpretation, change detection (CD) has been extensively studied by scholars in recent years. Affected by the illumination and the environment during bitemporal images' acquisition, there will be many pseudochanges, and the pseudochanges will seriously affect the effect of CD. Based on this, we propose a CD model named HMCNet, which introduces multilayer perceptron (MLP) into a convolutional neural network (CNN)-based CD model to form an MLP-CNN hybrid model. HMCNet has both the good feature extraction of CNN and the long-term dependence modeling ability of MLP, which can effectively overcome the interference of pseudochanges. In addition, the proposed cross-axis attention MLP can induce window attention of local features through shifted windows and, at the same time, form global attention to features through the interaction between information flows on the cross-axis, which effectively improves the comprehensive performance of MLP block. Extensive experiments on three public benchmark datasets show that HMCNet can achieve better performance with fewer parameters and Flops, and still maintain good generalization ability with fewer train data.**

***Index Terms*— Change detection (CD), cross-axis attention, hybrid model, multilayer perceptron (MLP), remote sensing (RS) images.**

I. Introduction

REMOTE sensing (RS) images change detection (CD) refers to a task that uses multiple RS images of the Earth's surface taken at the same location but at different times to assist in analyzing the change information of ground objects. As an important task in the field of RS image interpretation, CD has received continuous attention and research in recent years. Using the change information of specific areas on the Earth's surface, some individuals and institutions can make accurate decisions, which can be extremely helpful for productivity and life in specific fields. At present, CD has been widely used in tasks such as land use monitoring [1], urban expansion [2], environmental monitoring [3], and disaster monitoring [4], [5]. With the rapid development of RS observation technology, people can acquire more high-resolution RS images for CD. Since manual interpretation and labeling of change areas require a large amount of a priori knowledge of professionals, which is time-consuming and laborious, therefore, there is an urgent need to develop automatic RS techniques to identify change areas quickly and accurately.

Traditional CD algorithms can be divided into two categories: one is pixel-based methods [6], [7], [8], [9] and the other is object-based methods [10], [11], [12]. The pixel-based CD methods generate the final change map based on the pixel differences between multitemporal or bitemporal images using a clustering algorithm [6], [8] or setting a suitable threshold [7]. This method has high requirements for image preprocessing, which usually goes through several steps such as radiation correction [13] and geometric correction [14]. Since previous RS images often have low resolution, leading to the possibility of containing feature information, such as houses, roads, and vegetation, within a single pixel point at the same time, pixel-based CD methods inevitably produce some erroneous isolated change pixels, at which point the method is no longer well suited for the job, and therefore, object-based CD methods have been developed. The object-based method first extracts and divides the semantic features of the multitemporal or bitemporal images, respectively, and then determines whether the area has changed according to the difference between the feature groups. The effect of this method is also highly related to the accuracy of image registration. Both of the above methods require personnel with professional knowledge to set appropriate thresholds to achieve good results. However, the quality of RS images collected in recent years has increased significantly. The traditional CD method cannot be well adapted to the detection of complex scene changes at high resolution due to its poor robustness.

Most of the current CD models lack good feature extraction ability and long-range dependence modeling ability, so there are still many problems in detecting effective changes and filtering out disturbing change information. In CD, pseudochange [15] is defined as disturbing change information caused by factors such as seasons or lighting during the shooting process. Fig. 1 shows some pseudochange and irrelevant-change interference problems that are often encountered in the CD process. The yellow box marks the pseudochange of the roof caused by the influence of lighting; the red box marks the pseudochange of the shadow caused by the light angle, which is a situation that is difficult for the network to judge; the green box marks the irrelevant

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Fig. 1. Examples of pseudochange and irrelevant-change.

changes caused by the movement of the vehicle. How to filter pseudochange interference information is a hot spot and an important challenge in the current CD research process.

Larger datasets and the continuous improvement of computer computing power have made a series of data-driven deep-learning methods popular in the field of computer vision. Among them, three kinds of deep neural networks represented by convolutional neural networks (CNNs) [16], vision transformers (ViTs) [17], and multilayer perceptrons (MLPs) [18] are more popular. In the following, we will individually introduce the specific application of each network in the CD field and elicit our work.

A. CNN-Based Models

CNN-based models are the most widely used in the field of computer vision, and some well-designed classical networks, such as AlexNet [16], VGGNet [19], and ResNet [20], have been widely used in tasks such as salient object detection [21], semantic segmentation [22], target tracking [23], cross-modal retrieval [24], and super-resolution reconstruction [25], and have achieved good results. Daudt et al. [26] proposed three FCN-based CD models, which present the ideas of early fusion, Siamese connection, and Siamese difference that have become paradigms in the field of CD. Due to the dual-input nature of the CD task, the single-branch structure generates spatial feature entanglement and noncorrespondence when extracting bitemporal image features, while the Siamese structure can effectively avoid these problems by extracting features from bitemporal images separately. Fang et al. [27] proposed a CNN-based CD model by nesting the Siamese structure into the UNet++ [28] network, which utilizes the dense skip connection mechanism to effectively extract features and further enhance them using the proposed ensemble channel attention module, thus achieving good results. Lu et al. [29] proposed a multiscale feature progressive fusion network based on the feature pyramid network, in which the multiscale feature aggregation and distribution modules enable the network to correctly detect change regions. Although these models have achieved some success, the inherent translation invariance and local connectivity of CNNs have limitations in segmentation tasks, such as CD, and CNNs also lack the ability to capture long-range dependencies between features, which will lead to the inability of CNN-based models to determine pseudochange when performing CD on complex scenes. Therefore, Transformer has been gradually introduced to the field of CD in recent years.

B. Transformer-Based Models

The Transformer was originally designed to solve natural language processing (NLP) tasks [30], [31] and has been widely used in machine translation, text mining, and other fields. Recently, Dosovitskiy et al. [32] introduced Transformer to the computer vision field, showing that image classification tasks can also be performed well with a pure Transformer. Since then, the application of Transformer to other visual fields has become a research hotspot in recent years. The application of Transformer in the field of CD begins with that Chen et al. [33] represented bitemporal images as several tokens and used Transformer to establish a long-range contextual relationship in a token-based space, which confirmed the feasibility of Transformer to handle CD tasks, but this model only replaced the last convolutional stage of ResNet18 with Transformer compared to the pure convolutional model. Zhang et al. [34] designed a pure Transformer model with a Siamese structure, which adopted an encoder–decoder structure and adopted the Swin Transformer as the basic unit at each stage, further exploring the potential of the pure Transformer structure in the CD field. The Transformer-based CD model proposed by Bandara et al. [35] uses a hierarchical Transformer as an encoder, with a feature difference module and a lightweight MLP decoder to achieve good results. However, due to the inherent nature of its structure, Transformer requires a larger amount of data and more computational consumption, which is very expensive in the current computing environment. Therefore, networks with MLP structures have been extensively studied recently.

C. MLP-Based Models

MLP has been reconsidered in recent years due to its simple structure and efficient computation [36], [37]. Compared to CNN and Transformer structured models, MLP-based models can introduce less induction bias [38] and save more computational effort, which is important for whatever computer vision task. Tolstikhin et al. [39] designed an architecture based on MLPs and demonstrated that networks without convolution and attention can also achieve good performance in vision tasks. Chen et al. [40] designed an MLP-based recurrent structure model, which can effectively capture the spatial relationship between features and achieve a better tradeoff between performance and efficiency. To the best of our knowledge, MLP-based models have not been introduced in the field of deep-learning-based CD, and most of the existing MLP-based CD methods are non-end-to-end traditional algorithms that require human intervention. Therefore, to introduce long-range dependence modeling capability and efficient computational efficiency into the CD domain, we implement an MLP-based end-to-end CD model for the first time. However, we found in

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our experiments that simple MLPs that have not been carefully designed are still slightly insufficient in feature extraction, so we consider redesigning the internal structure of MLPs and adopting a hybrid model scheme, because this largely avoids the negative effects of repeated stacking of MLP blocks.

D. Potential of Hybrid Models

CNN has been used until now due to its powerful feature extraction ability. Its parameter sharing feature, combined with the Siamese structure, can not only save the number of parameters when sharing weights but also makes the model easier to establish feature associations between bitemporal images. Therefore, the Siamese structure is used in most of the CD models nowadays. However, due to the local perception of CNN, although it performs well in processing image local content associations in any dimension, it is not good at modeling long-range dependencies between image features. Therefore, it is difficult for pure CNN models to discriminate pseudochange when dealing with high-resolution complex scenes, resulting in worse CD results. In view of this, we introduce MLP into the CNN model to replace several convolution stages in it to form a hybrid model of MLP-CNN, so as to make up for the lack of the long-range dependence modeling ability of the pure CNN model. Our proposed HMCNet uses a Siamese structured UNet nested in the encoding stage as the main body and uses our designed convolutional block and cross-axial attention MLP block to replace stage1-3 and stage4-5 of the Siamese UNet, respectively. The convolution block and the MLP block can provide different representations for the model at the feature level so that the two types of features can achieve complementary effects and ultimately improve the CD capability of the model. Finally, most models are designed to be more complex and more computationally intensive in order to obtain better results, which will make the model less dominant in inference speed and parameter quantity, and are not conducive to fast CD. We considered this problem at the beginning of the design, so our model achieves good performance while being lightweight.

The main contributions of this article are given as follows.

1) We propose a hybrid end-to-end RS model based on CNN and MLP, which takes into account the powerful feature extraction capability of CNN and the long-range dependence modeling capability of MLP. The feature representations provided by the two types of basic blocks achieve complementary enhancement, which can effectively eliminate the impact caused by pseudochange at CD in complex scenes.
2) With almost no extra computation compared to regular MLP, our proposed cross-axial attention MLP utilizes the cross-axial shifted strategy to fully interact with local features at the channel level to achieve the effect of modeling global long-range dependencies, effectively compensating for the shortcomings of pure CNN networks.
3) We conducted sufficient experiments on three publicly benchmark CD datasets (CDDs), and the quantitative results and visualization results show that our model outperforms recent state-of-the-art (SOTA) models. When the quantitative results are comparable to or better than other models, our model has great advantages in terms of parameters and complexity, and outperforms almost all deep-learning-based end-to-end CD models in inference speed.

TABLE I
CONFIGURATIONS OF FOUR DIFFERENT SCALES OF HMCNET

Model	C1	C2	C3	C4	C5	Params.(M)	FLOPs(G)
HMCNet/8	8	16	32	64	128	0.55	0.70
HMCNet/16	16	32	64	128	256	2.19	2.69
HMCNet/32	32	64	128	256	512	8.73	10.53
HMCNet/48	48	96	192	384	768	19.61	23.54

The remainder of this article is structured as follows. Section II describes our method in detail, Section III presents the experiments and gives a detailed analysis of the experimental results, and Section IV discusses the impact of network structure settings and problems in the dataset on the experimental results. Finally, the conclusions of this article are drawn in Section V.

II. METHODOLOGY

In this section, we first introduce the overall architecture of our proposed network, followed by the residual convolution block and the cross-axis attention MLP block, respectively, and, finally, the loss function used.

A. Overall Network Architecture

The proposed HMCNet is a typical codec structured network, in which we have transformed the encoder part into a Siamese structure to facilitate feature extraction and condensation of bitemporal images. The stage1-3 of the network uses CNN, and the stage4-5 uses MLP, which gives the network a good feature extraction and long-range dependence modeling capability, and the skip connection mechanism enables information between the same modalities to fully interact. The overall architecture of the network is shown in Fig. 2. Instead of using the conventional interpolation operation, we use transposed convolution for upsampling. We believe that transposed convolution is different from the "hard" upsampling of interpolation, which has learnable upsampling parameters. C1, C2, C3, C4, and C5 represent the channel hyperparameters of the network. By controlling this set of hyperparameters, we can obtain models of different scales. We have applied four scales of HMCNet, the specific configuration of which is shown in Table I.

B. Residual Convolution Block

The CNN block in Fig. 2 shows the internal structure of the residual convolution block that we adopted. In order to prevent the gradient from disappearing or exploding during the training process, we introduce the residual structure into

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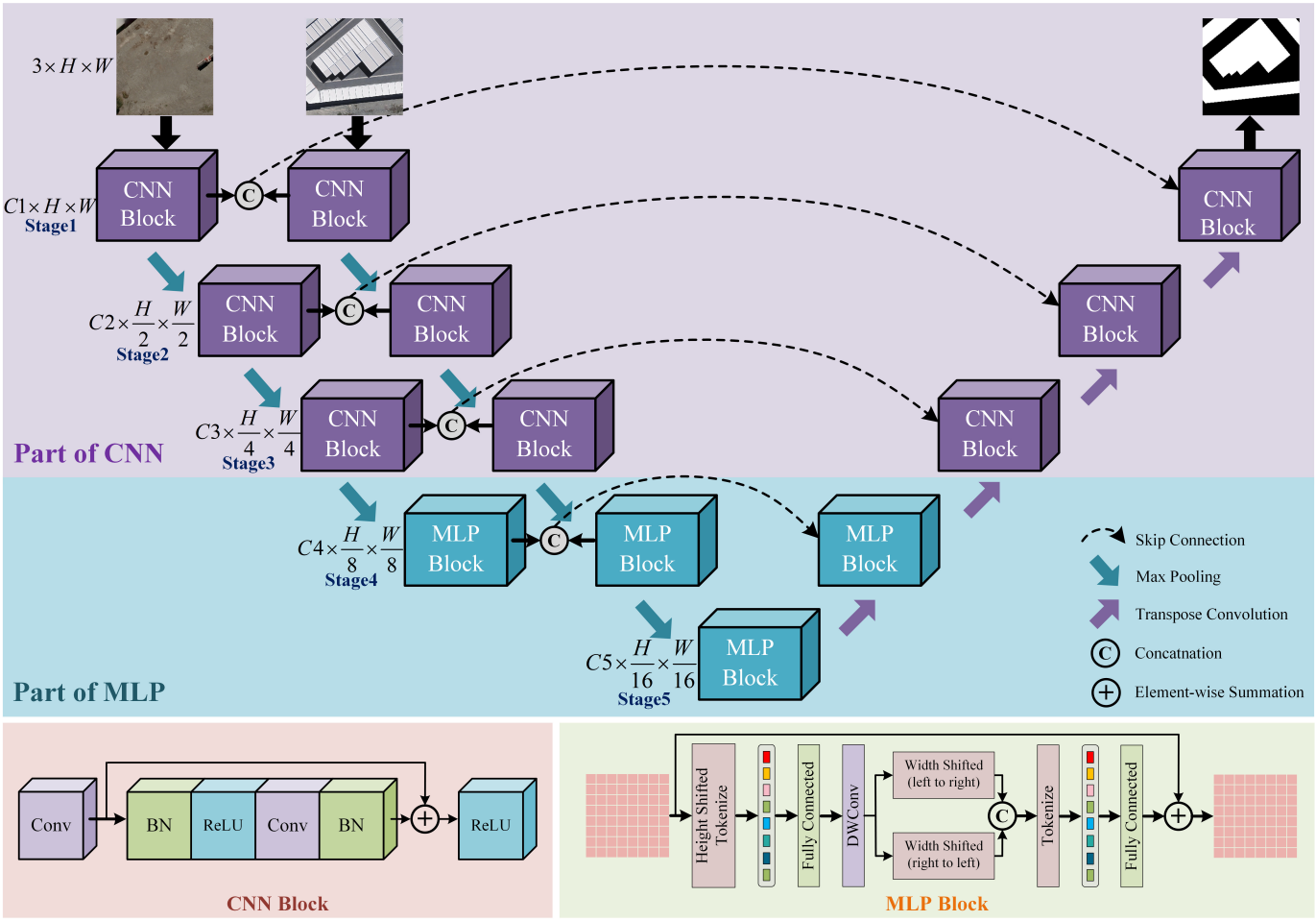


Fig. 2. Overall structure of the proposed HMCNet.

the convolution block to help stabilize the training process. The kernel size of the two convolution layers included in the CNN block is 3 × 3, the stride is 1, and the padding is 1, so the feature does not change the size after the CNN block processing. For the downsampling of the encoder, we use a max pooling operation with a kernel size of 2 × 2 and a stride of 2.

C. Cross-Axial Attention MLP Block

The internal structure of the cross-axial attention MLP block is shown in the MLP block in Fig. 2. Overall, the structure of the cross-axis attention MLP is a three-layer network formed by inserting a depthwise convolution (DWConv) layer between two layers of shifted fully connected networks. The first shifted fully connected network shifts the input features in height; the shift operation is similar to the sliding window operation of the Swin Transformer. We adaptively divide the features into windows from top to bottom in the channel dimension. This operation will introduce window-based attention, and the globally-attentive MLP is thus endowed with local attention capabilities, then project features into tokens and sends to the fully connected network. Then, the feature will pass through a layer of DWConv with a kernel size of 3, a stride of 1, and a padding of 1; groups are always consistent with the channel dimension of the current input feature. DWConv can encode the location information of the MLP feature [41] and effectively improve the performance of the MLP block. The second shifted fully connected network moves the features in the two width directions; the shift operation is consistent with the height shift operation, then concatenates the two width shifted features, projects them into tokens, and sends them to the fully connected network. At the same time, we also added a residual structure to the MLP block, which helps to stabilize the training and prevent the loss of effective information. Fig. 3 shows the process of the cross-axial shift operation. Assuming that the current input feature dimension is $F \in \mathbb{R}^{3\times5\times5}$, the shift operation adaptively determines the size and step of the shift window based on the dimensional information of the current feature (the default step is smaller than the window size), and then, the shift window samples the feature at height along the channel dimension, with the part outside the window padded with zero (gray blocks). We make the network focus on different information flows in the two cross-axis directions by axially shifting the input features at the channel level. The window formed by the axial shift can make the network pay more attention to the local features and form attention to the local features. Since the windows are deliberately designed to partially overlap, the local window attention in the two cross-axis directions can interact with each other to form global attention on features.

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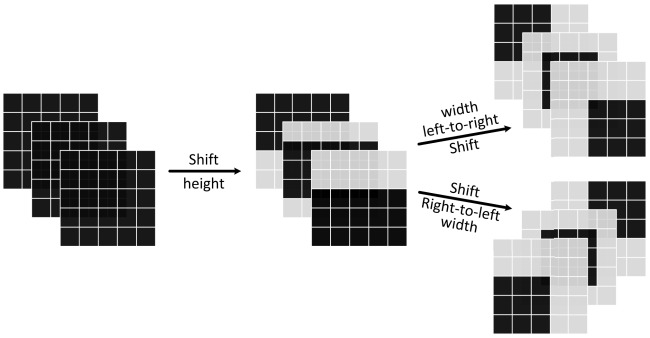


Fig. 3. Cross-axial shift operation.

The calculation process of the MLP block is given as follows:

$$T_W = \text{Tokenize}(\text{Shift}_W(X_i)) \tag{1}$$

$$X_m = \text{DWConv}(\text{FC}(T_W)) \tag{2}$$

$$T_H = \text{Tokenize}\big(\text{Concat}\big[\text{Shift}_{H_{l\to r}}(X_m), \text{Shift}_{H_{r\to l}}(X_m)\big]\big) \tag{3}$$

$$X_o = X_i + \text{FC}(T_H) \tag{4}$$

where X_i, X_m, and X_o represent input, middle, and output features, respectively, H represents the height, W represents the width, FC represents the fully connected network, DWConv represents the depthwise convolution, and Concat represents the concatenation of features at the channel level.

D. Loss Function

CD is a typical binary segmentation task, and the most used in such tasks is cross-entropy loss. However, since the number of two types of pixels in the CD task is often extremely unbalanced, we introduce the weighted cross-entropy loss used in [27]. To describe this loss, we can consider the predicted change map $\hat{Y}$ as a set of pixel points

$$\hat{Y} = \{\hat{y}_k, k = 1, 2, \ldots, H \times W\} \tag{5}$$

where H and W represent the size of the change map. The weighted cross-entropy loss is defined as follows:

$$L_{\text{wce}} = \frac{1}{H \times W} \sum_{k=1}^{H\times W} \text{weight[class]} \cdot \left(\log\left(\frac{\exp(\hat{y}[k][\text{class}])}{\exp(\hat{y}[k][0]) + \exp(\hat{y}[k][1])}\right)\right). \tag{6}$$

The value of "class" can only be changed and unchanged pixels, corresponding to 1 and 0, respectively. At the same time, we also introduce the dice loss

$$L_{\text{dice}} = 1 - \frac{2 \cdot Y \cdot \text{softmax}\big(\hat{Y}\big)}{Y + \text{softmax}\big(\hat{Y}\big)} \tag{7}$$

where Y represents the ground truth (GT). Finally, the mixed loss function L is defined as

$$L = L_{\text{wce}} + L_{\text{dice}}. \tag{8}$$

III. Experiments and Results Analysis

To fully evaluate the performance of our model, we conducted experiments on the CDD, Building CDD (BCDD), and LEarning, VIsion and RS-CD (LEVIR-CD), three publicly available benchmark datasets, and the experimental results show that our model outperforms the recent SOTA model. In this section, we first briefly introduce the adopted datasets and comparison methods, then introduce the evaluation metrics and implementation details used in the experiments, followed by a detailed analysis of the quantitative and visualization results on the three datasets, and, finally, introduce the ablation experiments.

A. Datasets and Preprocessing

1) BCDD [42]: This dataset contains a pair of RS images of size of 32 507 × 15 354 pixels collected in Christchurch, New Zealand, with a space resolution of 0.3 m/pixel. The area where the images were collected experienced an earthquake in February 2011, and a large number of large sparse buildings were rapidly reconstructed in the following years. Since the original dataset did not provide a standard segmentation scheme, we divided the images into 7434 pairs according to the principle of nonoverlapping and, finally, randomly divided these images into train, validation, and test sets in a ratio of 8:1:1.

2) CDD [43]: This dataset is a set of RS images acquired on Google Earth (digital Earth) that varies with seasons in the same area, including seven pairs of images with a size of 4725 × 2700 pixels and four pairs of images with a size of 1900 × 1000 pixels. The spatial resolution of the acquired images is 3–100 cm/pixel. The original dataset contains 16 000 pairs of images of size 256 × 256 pixels formed by cropping and rotation, of which 10 000 pairs are used for the train set, 3000 pairs are used for the validation set, and 3000 pairs are used for the test set.

3) LEVIR-CD [44]: This dataset contains 637 pairs of images of size 1024 × 1024 pixels collected from Google Earth with a spatial resolution of 0.5 m/pixel. Collected from 20 different regions in Texas, USA, these images contain many pseudochanges caused by seasons and lighting. The type of change is dominated by dense small buildings but also covers road construction, building decline, and so on. Considering the limitation of GPU memory during the experiment, we crop the images to a size of 256 × 256 pixels without overlapping, and randomly assign 7120 pairs of images as the train set, 2048 pairs as the validation set, and 1024 pairs as the test set.

Example graphs of changes in the three datasets are shown in Fig. 4. Because the three datasets focus on different types of changes, BCDD mainly focuses on the changes of large sparse buildings, CDD dataset mainly focuses on the changes of irregular roads and buildings, and LEVIR-CD mainly focuses on the changes of small dense buildings, the experiments on three datasets can comprehensively evaluate the performance of the model. We list the information of the three datasets in detail in Table II, in which the disparity in the ratio of changed pixels and unchanged pixels during training is also a big ordeal for the model.

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TABLE II
INFORMATION ON THE MAIN PARAMETERS OF THE THREE DATASETS THAT WE USED

Datasets	Spatial Resolution	Size	Number of Pixels			Number of Samples		
			Changed	Unchanged	Ratio	Train	Val	Test
BCDD	0.3m/pixel	256×256	21,352,815	477,759,663	1 : 22.37	5948	743	743
CDD	0.03-1m/pixel	256×256	134,068,750	914,376,178	1 : 6.82	10,000	3,000	3,000
LEVIR-CD	0.5m/pixel	256×256	30,913,975	637,028,937	1 : 20.61	7120	1024	2048

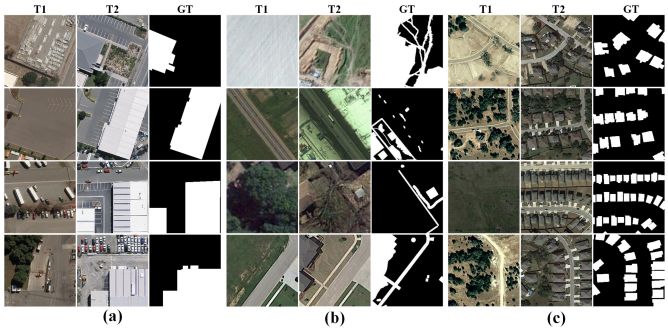


Fig. 4. Samples from three datasets, where T1 and T2 represent bitemporal images and GT represents GT. (a) BCDD. (b) CDD. (c) LEVIR-CD.

B. Comparison Methods

1) FC-EF [26]: A single-stream CD model based on UNet, where the bitemporal images are concatenated at the channel level at the model input and fed into the model for processing. The overall structure follows the UNet codec structure.

2) FC-Siam-conc [26]: It is a variant of FC-EF. The main improvement is to transform the coding part of the network into a Siamese structure, where two encoder branches process two images separately, and two features from the same level of the encoder are concatenated at the channel level and sent to the corresponding level of decoder for processing.

3) FC-Siam-diff [26]: The same network structure as FC-Siam-conc, with the difference that the feature fed into the decoder is the difference feature obtained by subtracting the two features generated by the encoder.

4) CDNet [45]: A codec structure model with skip connections. It adopts a single-stream structure, including four shrinking blocks and four expanding blocks. The convolution with a kernel size of 7 is uniformly used, and the number of channels in the middle layer of the network is always kept as 64.

5) STANet [44]: In the feature extraction part, ResNet18 with the Siamese structure is used to extract bitemporal features, and then, two sets of features are sent to the spatiotemporal attention module to model the spatiotemporal context relationship between features, and finally, the metric module generates the final change map.

6) BIT-CD [33]: The first to introduce Transformer into the CD field to form a CNN-Transformer hybrid model. A bitemporal image transformer is proposed to effectively simulate the long-range relationship between features, replacing the last stage of ResNet18 with BIT and, finally, feeding the features into the prediction head to generate a change map.

7) SNUNet/48 [27]: UNet++ is used as the backbone network, dense skip connections are used to reduce the loss of deep localization information, and the four-level features from the backbone network are sent to the ensemble channel attention module to refine the features. The initial channel number of this model is 48, which is the best performing model in [27].

8) ChangeFormer [35]: A Transformer-based CD network, which combines the hierarchical Transformer encoder and the MLP decoder in the Siamese network architecture. The Transformer encoder is responsible for extracting rough features, and the features on the Siamese encoder are processed by the difference module and sent to the MLP decoder output prediction.

The principle of our selection of the comparison method is to try to select the latest open-source model based on the principle of reproducibility, and the selected methods are the more classic and representative methods in the field of CD. Due to the randomness of the dataset division, in order to ensure the fairness of the experiment, we reexperimented all the comparison methods on three datasets, all experiments were conducted in the same hardware and software environment, and the settings for the comparative experiments were consistent with those described in the original paper, using the same datasets as those used in our model.

C. Evaluation Metrics

To compare the performance of our model with other models, we use F1 and IoU as the main quantitative evaluation metrics, and we additionally use three evaluation metrics of precision, recall, and overall accuracy (OA). The range of quantitative metrics is from 0 to 1, and the higher the value, the better the performance. The expressions of the five quantitative metrics are given as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{9}$$

$$\text{Rcall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{10}$$

$$\text{F1} = \frac{2\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{12}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{13}$$

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TABLE III
QUANTITATIVE RESULTS ON THE BCDD DATASET

Model	Params. (M)	Flops (G)	BCDD				
			Precision	Recall	F1	IoU	OA
FC-EF	1.29	2.92	82.28	70.66	76.03	61.33	97.92
FC-Siam-conc	1.93	4.55	40.09	73.84	51.97	35.11	93.63
FC-Siam-diff	1.75	3.99	38.82	71.80	50.40	33.69	93.40
CDNet	1.36	21.52	92.16	83.18	87.44	77.68	98.88
STANet	16.93	6.58	91.25	86.18	88.64	79.61	98.97
BIT-CD	3.55	10.6	86.07	85.61	85.84	75.19	98.68
SNUNet/48	28.34	97.87	88.35	87.80	88.07	78.69	98.89
ChangeFormer	267.90	129.27	93.44	85.70	89.40	80.84	99.12
HMCNet/8	0.55	0.70	90.58	88.76	89.66	81.25	99.04
HMCNet/16	2.19	2.69	94.18	87.65	90.80	83.15	99.17
HMCNet/32	8.73	10.53	92.42	89.95	91.17	83.77	99.19
HMCNet/48	19.61	23.54	92.27	92.74	92.51	86.06	99.30

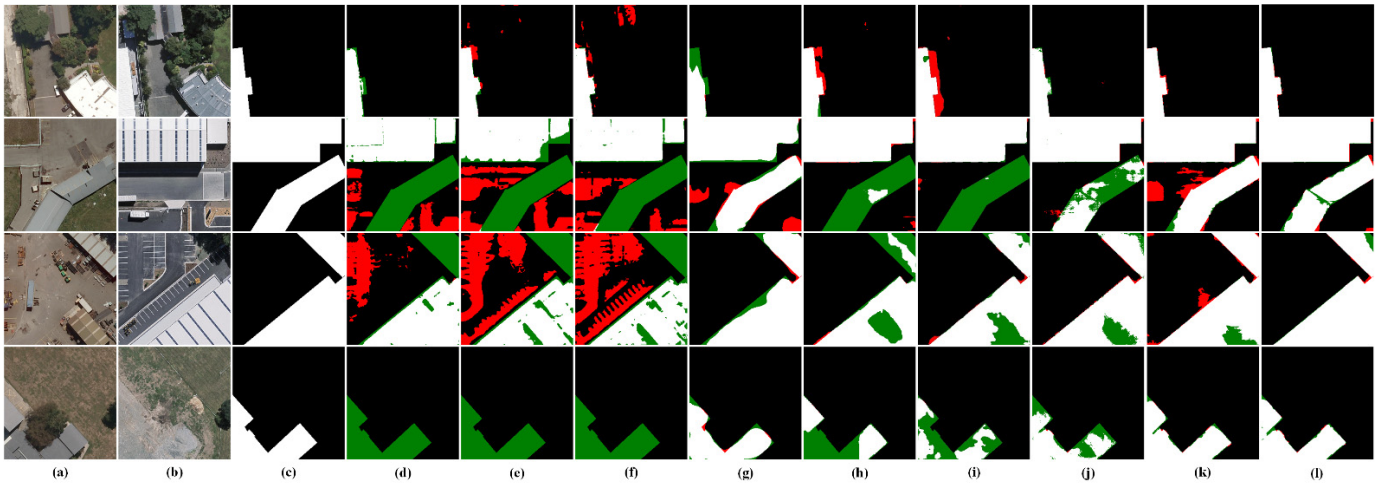


Fig. 5. Visualization results on the BCDD dataset. (a) T1 images. (b) T2 images. (c) GT. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) CDNet. (h) STANet. (i) BIT-CD. (j) SNUNet. (k) ChangeFormer. (l) HMCNet/48.

where TP is true positive, representing correctly predicted changed pixels; TN is true negative, representing correctly predicted unchanged pixels; FP is false positive, representing incorrectly predicted changed pixels; and FN is false negative, representing incorrectly predicted unchanged pixels. At the same time, the size of model parameters and computational complexity are also listed to comprehensively evaluate the balance of the model.

D. Implementation Details

We trained our model using the PyTorch framework on a single NVIDIA TITAN RTX (24 GB) GPU, the batch size is set to 16, and the initial learning rate is set to 0.0001, using the AdamW optimizer with exponential learning rate decay. The model was trained on all datasets with 300 epochs, and the learning rate warm-up operation was performed in the first 5 epochs to mitigate the instability of the model during the initial training phase.

E. Comparative Experiments

1) Quantitative Analysis on BCDD: Table III lists the performance results of our model and other comparative models on the BCDD dataset. The results show that HMCNet/48 significantly outperforms other comparison models on all metrics, and the F1 and IoU metrics are 3.87% and 6.45% higher than the second best STANet. Our smallest model HMCNet/8 also outperforms all comparison models on the main metrics F1 and IoU. Compared with the SOTA Transformer-based CD model, ChangeFormer, our HMCNet/8 outperforms ChangeFormer on the BCDD dataset with only 0.2% of its parameters and 0.5% of its computation.

2) Visualization Analysis on BCDD: Fig. 5 shows the visualization results of our model and other comparison models

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TABLE IV
QUANTITATIVE RESULTS ON THE CDD DATASET

Model	Params. (M)	Flops (G)	CDD				
			Precision	Recall	F1	IoU	OA
FC-EF	1.29	2.92	76.56	43.49	55.47	38.38	91.76
FC-Siam-conc	1.93	4.55	88.00	53.58	66.61	49.93	93.66
FC-Siam-diff	1.75	3.99	88.49	51.53	65.14	48.30	93.49
CDNet	1.36	21.52	91.93	84.87	88.26	78.98	97.34
STANet	16.93	6.58	88.97	94.31	91.56	84.44	97.95
BIT-CD	3.55	10.6	95.86	94.59	95.22	90.88	98.88
SNUNet/48	28.34	97.87	96.82	96.72	96.77	93.74	99.24
ChangeFormer	267.90	129.27	95.28	93.83	94.55	89.66	98.72
HMCNet/8	0.55	0.70	93.68	90.06	91.84	84.90	98.11
HMCNet/16	2.19	2.69	95.66	95.01	95.33	91.08	98.90
HMCNet/32	8.73	10.53	96.95	97.31	97.13	94.41	99.32
HMCNet/48	19.61	23.54	97.45	97.75	97.60	95.31	99.43

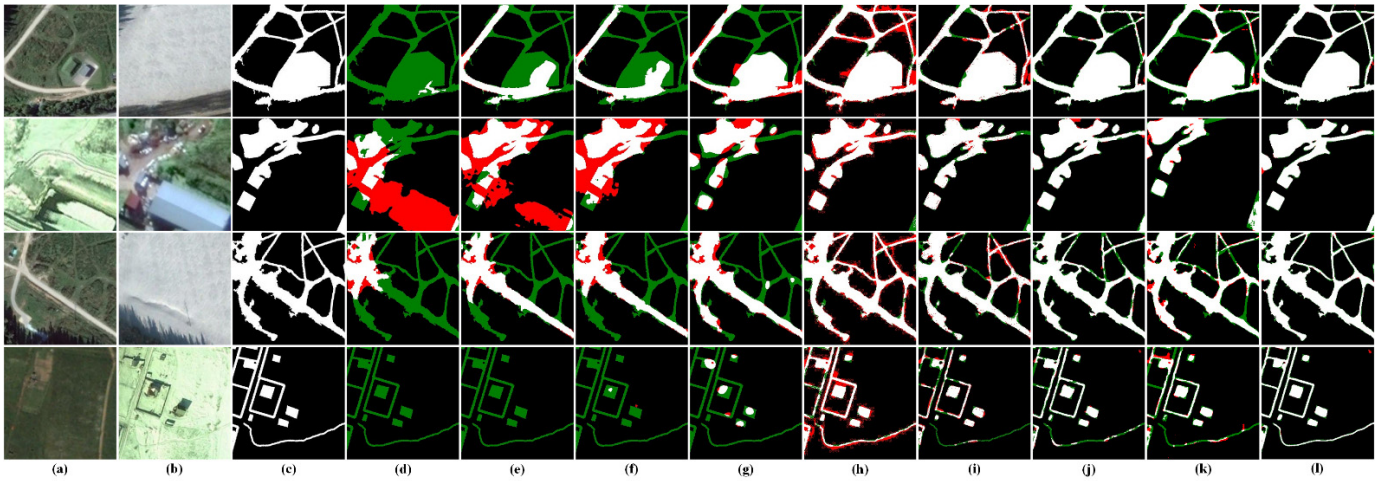


Fig. 6. Visualization results on the CDD dataset. (a) T1 images. (b) T2 images. (c) GT. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) CDNet. (h) STANet. (i) BIT-CD. (j) SNUNet. (k) ChangeFormer. (l) HMCNet/48.

on the BCDD dataset. For the convenience of observation, we mark the pixel positions of missed detection in green and the pixel positions of false detection in red. The results in the second row show that, due to the small difference between the building and the ground in the figure, almost all of the comparison models, except our model, did not detect the complete change in the building's outline. In the fourth row, due to the shadow of the tree, many comparison models fail to detect the real contour of the changing building. Although the contour detected by ChangeFormer is relatively complete, it is obtained under the condition of a larger model scale and more complex calculation. Compared with ChangeFormer, our model achieves the same detection performance with 7.3% of the parameters and 18.2% of the computation.

3) Quantitative Analysis on CDD: Table IV lists the performance results of our model and other comparative models on the CDD dataset. The results show that HMCNet/48 outperforms the CNN-based SOTA model SNUNet/48 by 0.83% and 1.57% on F1 and IoU metrics, respectively. It is worth mentioning that our smallest model HMCNet/8 achieves comparable performance with only 3.2% of STANet's parameters and 10.6% of the Flops. From the results of BIT-CD and HMCNet/8, it can be seen that the model with a smaller model capacity is difficult to deal with the detection of irregular objects in complex scenes.

4) Visualization Analysis on CDD: Fig. 6 shows the visualization results of our model and other comparison models. For the convenience of observation, we mark the pixel positions of missed detection in green and the pixel positions of false detection in red. On the whole, all the comparison models can hardly detect the changes in small roads well and cannot achieve smooth division in contour division. The visualization results of our model are almost indistinguishable from those of GT and are almost identical to GT in small road detection and contour division.

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TABLE V
QUANTITATIVE RESULTS ON THE LEVIR-CDD

Model	Params. (M)	Flops (G)	LEVIR-CD				
			Precision	Recall	F1	IoU	OA
FC-EF	1.29	2.92	82.27	66.28	73.41	58.00	97.55
FC-Siam-conc	1.93	4.55	86.81	67.66	76.05	61.36	97.83
FC-Siam-diff	1.75	3.99	86.55	74.38	80.00	66.68	98.11
CDNet	1.36	21.52	88.38	85.08	86.70	76.52	98.67
STANet	16.93	6.58	80.99	91.21	85.79	75.12	98.46
BIT-CD	3.55	10.6	91.95	88.57	90.23	82.19	99.02
SNUNet/48	28.34	97.87	91.66	88.48	90.04	81.89	99.00
ChangeFormer	267.90	129.27	91.53	88.86	90.17	82.10	99.01
HMCNet/8	0.55	0.70	86.55	87.07	86.81	76.70	98.65
HMCNet/16	2.19	2.69	89.74	88.21	88.97	80.13	98.89
HMCNet/32	8.73	10.53	90.81	89.90	90.35	82.40	99.02
HMCNet/48	19.61	23.54	91.68	89.82	90.74	83.05	99.07

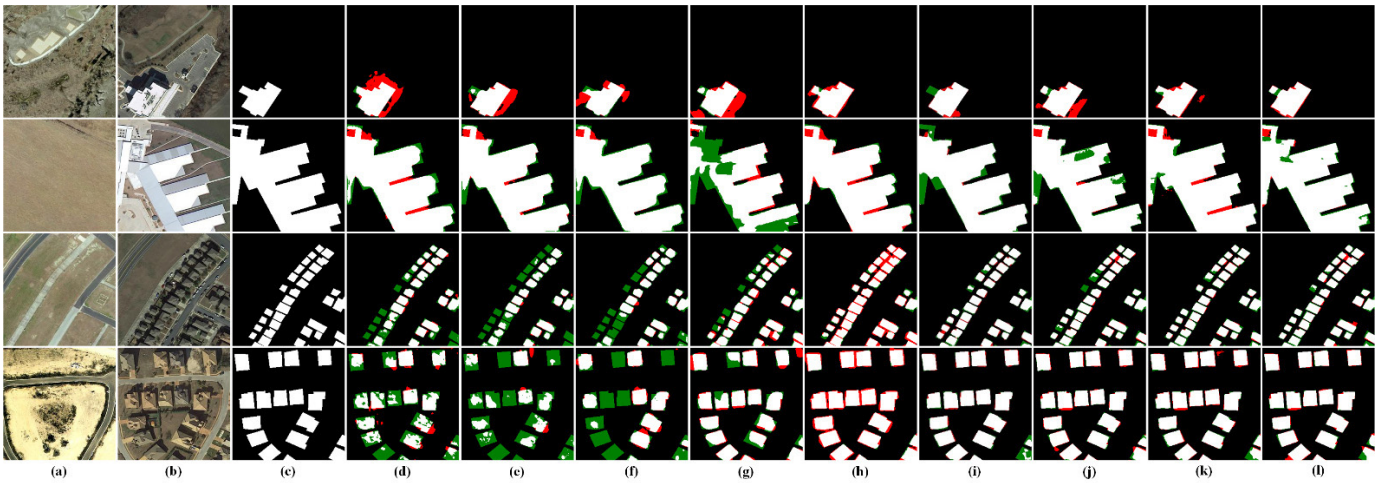


Fig. 7. Visualization results on the LEVIR-CDD. (a) T1 images. (b) T2 images. (c) GT. (d) FC-EF. (e) FC-Siam-conc. (f) FC-Siam-diff. (g) CDNet. (h) STANet. (i) BIT-CD. (j) SNUNet. (k) ChangeFormer. (l) HMCNet/48.

5) Quantitative Analysis on LEVIR-CD: Table V lists the performance results of our model and other comparative models on the LEVIR-CDD. The results show that the F1 and IoU metrics of HMCNet/48 reach 90.74% and 83.05%, which are 0.51% and 0.86% higher than BIT-CD, respectively. In addition, our HMCNet/32 has already surpassed all the comparison methods in the main indicators, F1 and IoU, and significantly outperformed the CNN-based model SNUNet/48 in the parameters and Flops indicators.

6) Visualization Analysis on LEVIR-CD: We list the visualization results of our model and other comparison models in Fig. 7. For the convenience of observation, we mark the pixel positions of missed detection in green and the pixel positions of false detection in red. It can be seen from the figure that FC-EF, FC-Siam-conc, and FC-Siam-diff have more missed detections and false detections in the face of slightly complex scenes; STANet does not delineate regular boundaries well for densely packed small houses; our model performs well in the face of large buildings, isolated small building, and dense small buildings; and the results are basically consistent with GT.

F. Ablation Experiments

In order to verify the impact of the CNN block and MLP block introduced in the design process of the proposed model on the performance of the model, we designed the following ablation experimental set.

1) *UNet:* The original UNet with 64 channels in the first stage.
2) *Conv Stage:* Replace the convolution block of the original UNet with our residual convolution block and change the encoder part to a Siamese structure.
3) *Conv Stage + MLP:* The fourth and fifth stages of the Conv Stage network are replaced by ordinary MLP

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TABLE VI
QUANTITATIVE RESULTS OF MODEL ABLATION EXPERIMENTS ON THE BCDD DATASET

Network	Params.(M)	Flops(G)	BCDD				
			Precision	Recall	F1	IoU	OA
UNet	42.71	77.57	59.21	88.40	70.92	54.95	96.62
Conv Stage	24.12	28.23	90.99	90.42	90.71	82.99	99.14
Conv Stage + MLP	18.57	23.36	90.23	91.29	90.76	83.08	99.13
Conv Stage + MLP w/ DWConv	18.59	23.37	92.86	89.34	91.07	83.60	99.18
Conv Stage + H-shifted MLP w/ DWConv	18.59	23.37	94.60	89.18	91.81	84.85	99.26
Conv Stage + W-shifted MLP w/ DWConv	18.59	23.37	92.92	90.22	91.55	84.42	99.22
Conv Stage + HW-shifted MLP w/ DWConv	18.59	23.37	92.58	92.11	92.35	85.78	99.29
Conv Stage + Cross-axis MLP w/ DWConv	18.59	23.37	92.27	92.74	92.51	86.06	99.30

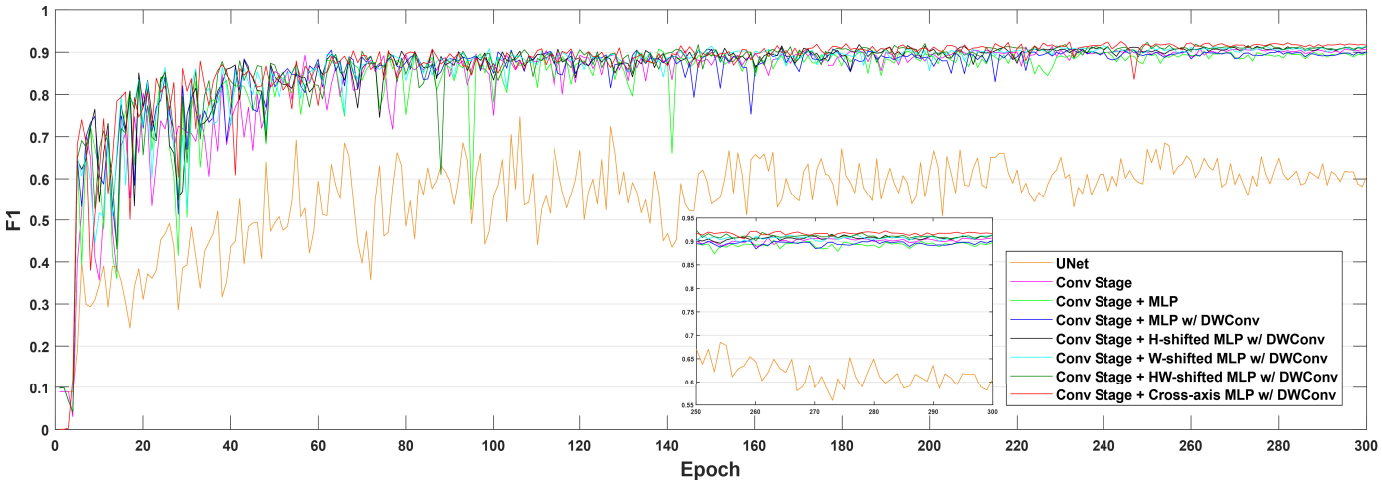


Fig. 8. Line chart of ablation experiments on the BCDD dataset; small windows show the details.

blocks, which only contains two layers of fully connected networks.

4) *Conv Stage + MLP w/ DWConv:* Add DWConv between two layers of fully connected networks.
5) *Conv Stage + H-shifted MLP w/ DWConv:* Height-shift operation is used before the first layer of the fully connected network.
6) *Conv Stage + W-shifted MLP w/ DWConv:* Width-shift operation is used before the first layer of the fully connected network.
7) *Conv Stage + HW-shifted MLP w/ DWConv:* A height-shift operation is used before the first-layer fully connected network, and a width-shift operation is used before the second-layer fully connected network to form a single-axis shifted MLP block.
8) *Conv Stage + Cross-axis MLP w/ DWConv:* Our model HMCNet/48 after adding all the improvements.

All experiments except UNet adopt the network with 48 channels in the first stage.

Table VI lists the results of ablation experiments; in order to more intuitively show the process of model "evolution," we show in Fig. 8 the line chart of the F1 indicator on the BCDD validation set during the training of these eight experiments. Because the original UNet is a single-stream structure, it is not suitable for the dual input requirements of CD tasks, so the metrics fluctuate greatly during the training process, and finally, the F1 only reached 70.92%; Conv Stage introduces a convolution block with residual structure, which helps to stabilize the data during the training process, and the Siamese structure of the encoder is more suitable for the CD task, so the metrics are greatly improved compared to UNet. Although Conv Stage + MLP w/o DWConv introduces MLP blocks to add the ability to model long-range relationships; due to the simple design and the lack of Conv layers to encode MLP features effectively location information, there are more fluctuations in the training process; and the main metrics F1 has almost no change compared with Conv Stage. In the next few experiments, due to the addition of DWConv to encode the position information of MLP features, the shifted MLP induces the local attention of blocks, so the training process is more stable than before, and the F1 indicator is gradually improved. From the third and fourth experimental comparisons in Table VI, it can be seen that the introduction of DWConv does not significantly increase the size of the model, and we also detail why we introduce DWConv here instead of the original convolution in Section IV. Subsequent sets of

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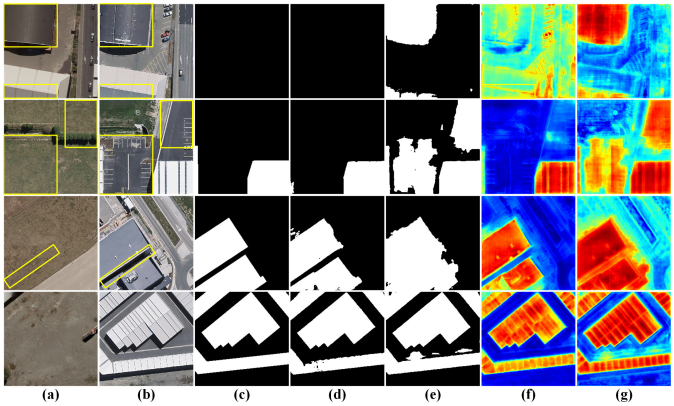


Fig. 9. Comparison of visualization results and heatmaps. (a) T1 images. (b) T2 images. (c) GT. (d) HMCNet/48. (e) UNet. (f) Heatmap of the last stage of HMCNet. (g) Heatmap of the last stage of UNet.

experiments confirm that the shift operation that we adopt introduced almost no additional parameters and Flops. Finally, our "complete body" model is 21.59% higher than UNet in the F1 metric, but the parameters and Flops are reduced by 56% and 70%, respectively.

To further illustrate the progress of our model in eliminating the effects of pseudochange, we present a heat map of UNet and HMCNet/48 on the BCDD dataset in Fig. 9. As can be seen from the figure, UNet misinterprets these pseudochanges as true when faced with roof color differences, ground differences, and shadows, but HMCNet/48 has good location information encoding and long-range relationship modeling capabilities, so it can easily identify these pseudochanges and reduce the false detection rate. However, we observed in our experiments that, in a few cases, HMCNet will misjudge a container as a house and a small house as a car, resulting in false detections and missed detections. Therefore, in the next step, we will focus on model optimization to reduce the interest of irrelevant changes and enhance the detection ability of small objects.

IV. Discussion

In this section, we present a comparison of the efficiency metrics of the CD models and further discuss the impact that dataset and model design may have on the final detection results.

A. Comparison of Model Comprehensive Performance

We have compared the comprehensive performance of HMCNet with other models, as shown in Fig. 10. The comparison is based on the three metrics of parameters, Flops, and inference time. The data of all models are obtained on a server equipped with an Intel Xeon Gold 5120 @ 2.20-GHz CPU and an NVIDIA TITAN RTX (24 GB) GPU. The inference time is based on the visualization time of a pair of 256 × 256 size images. The F1 value of the ordinate is the performance of the model on the BCDD dataset. As can be seen from the figure, our model is relatively optimal in terms of comprehensive performance, using fewer parameters and Flops to obtain better performance, while also leading in inference time by a large margin. Overall, our model achieves superior performance with less complexity and faster inference speed compared to other SOTA CD models. The comprehensive performance indicators of all models are shown in Table VII.

TABLE VII
Efficiency Metrics for HMCNet and Other Models

Model	Params. (M)	Flops (G)	Infer. Time (ms)	BCDD F1
UNet/32	10.69	19.50	28.69	74.73
UNet/64	42.71	77.57	45.22	84.84
FC-EF	1.29	2.92	24.10	76.03
FC-Siam-conc	1.93	4.55	29.91	51.97
FC-Siam-diff	1.75	3.99	27.71	50.40
CDNet	1.36	21.52	14.44	87.44
STANet	16.93	6.58	40.40	88.64
BIT-CD	3.55	10.6	28.82	85.84
SNUNet/48	28.34	97.87	45.67	88.07
HMCNet/8	0.55	0.70	15.09	89.66
HMCNet/16	2.19	2.69	18.25	90.80
HMCNet/32	8.73	10.53	19.21	91.17
HMCNet/48	19.61	23.54	23.63	92.51

TABLE VIII
Quantitative Results on the CDD Dataset

Model	CDD				
	Precision	Recall	F1	IoU	OA
HMCNet/48(20%)	94.84	93.49	94.16	88.97	98.63
HMCNet/48(50%)	97.13	97.03	97.08	94.33	99.31
HMCNet/48(80%)	97.50	97.91	97.70	95.51	99.46
HMCNet/48(100%)	97.45	97.75	97.60	95.31	99.43

B. CDD Dataset Overlap Problem

Compared with the other two datasets that we used, the CDD dataset has more diverse change types and finer annotations, but many models generally perform better on the CDD dataset than on other "simple" datasets, so we conduct a detailed observational study on the train and test data of the CDD dataset. As shown in Fig. 11, we observed that there is a lot of overlap in the data of the train set and the test set. Many images are used as a new set of data after a simple shift and rotation, and a large number of test images and train images have a high degree of overlap. Therefore, this dataset cannot effectively detect the generalization ability of the model but rather test the "memory" ability of the model. In response to this problem, we designed a set of experiments to test the generalization ability of HMCNet/48 on the CDD dataset, using 80%, 50%, and 20% of the original train set size (10 000 images) to train the model and then test the model on the test set. Table VIII lists the experimental results. There are two points worth noting. The first point is that,

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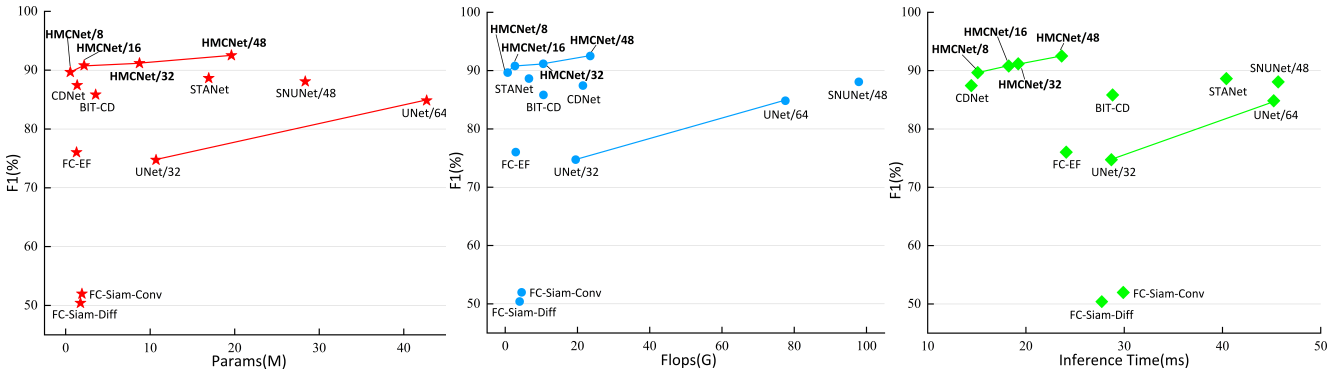


Fig. 10. Floating point plot of efficiency metrics for HMCNet and other CD models.

TABLE IX
QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THE NUMBER OF MLP BLOCKS ON THE BCDD DATASET

	Params.(M)	Flops(G)	BCDD				
			Precision	Recall	F1	IoU	OA
MLP-45	19.61	23.54	92.27	92.74	92.51	86.06	99.30
MLP-345	19.22	20.38	93.73	89.62	91.63	84.55	99.24
MLP-2345	19.14	17.22	93.09	89.45	91.23	83.88	99.20
MLP-12345	19.12	16.76	91.86	90.10	90.97	83.44	99.17

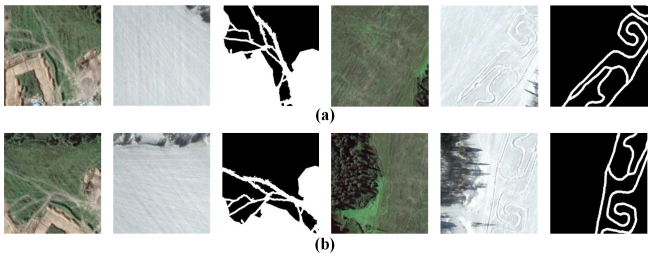


Fig. 11. Examples of data overlap problems for the CDD dataset. (a) Train set. (b) Test set.

when the size of the train set is 80% of the original train set, the model shows better performance. We infer that excessive content emphasis due to repeated data limits the generalization ability of the model [46]. The second point is that, when the train data (2000 images) is significantly less than the test data (3000 images), HMCNet/48 still performs better than most of the comparison models, which fully shows that our model has excellent generalization ability.

C. Impact of MLP Block Number

The local perception capability of CNNs has proven to be a useful inductive bias in many tasks, and therefore, introducing a CNN backbone into the early stages of the ViT structure can significantly improve the performance of the model [47]. Inspired by this, we introduced the designed cross-axial attention MLP into the pure CNN network, and to determine the optimal configuration of the number of MLP blocks in the model, we performed a set of experiments, as shown in Table IX MLP-45 means that the original convolutional blocks were replaced with MLP blocks in the fourth and fifth stages of the network. Since MLP-5 replaced only one convolutional block and lacked the interaction between codec information, we did not adopt this experiment. At the same time, we draw the F1 line graph of the network on the BCDD validation set in Fig. 12. As can be seen from the figure, as the number of MLP blocks increases, the performance of the network gradually decreases, which is due to the fact that the feature extraction ability of MLP in the early stage of the network is not as good as that of CNN. Therefore, our current design mix fully balances the model's local perception and long-range dependence modeling capabilities.

D. DWConv Versus Ordinary Conv

We added DWConv between the two-layer fully connected network of the MLP block, which helps to encode the location information of MLP features and further enhances the performance of the MLP block. As shown in Table X, we compared the models using DWConv and Ordinary Conv on three datasets, and the only difference between the two convolutions is that the groups of DWConv are always the same as the number of channels of the input feature, while the group of Ordinary Conv is always 1, the results show that the Ordinary Conv does outperform the DWConv on the BCDD and CDD datasets, which is due to the smaller number of parameters of DWConv, and the model capacity is also smaller. However, compared with DWConv, the parameters of Ordinary Conv increased by 46.7% and Flops increased by

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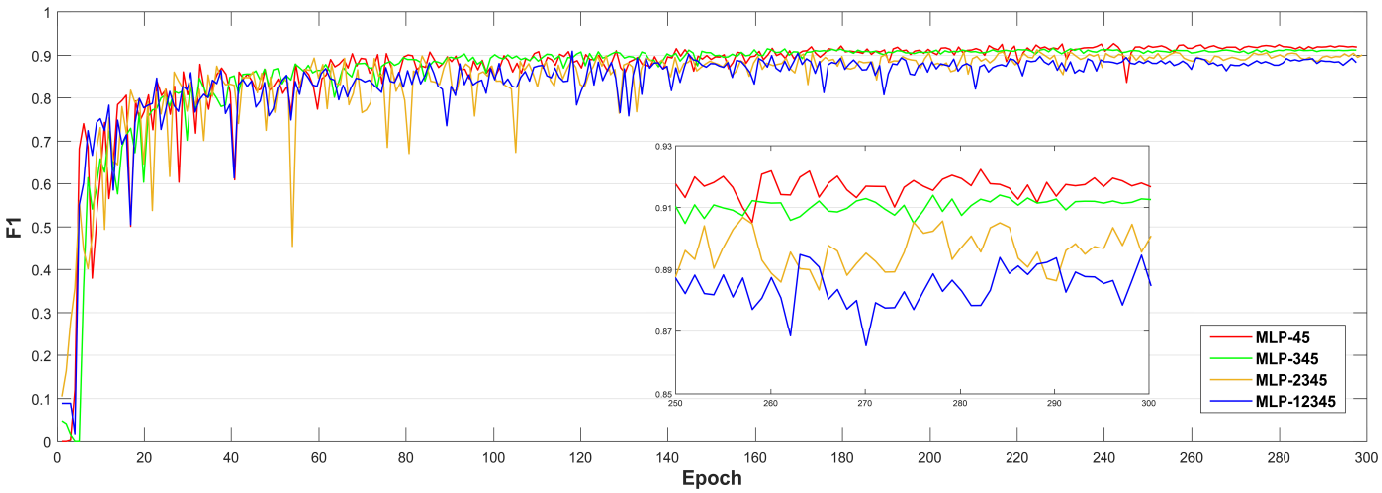


Fig. 12. Floating point plot of efficiency metrics for HMCNet and other CD models.

TABLE X
QUANTITATIVE RESULTS ON THREE DATASETS USING DIFFERENT CONVOLUTIONS

	Params.(M)	Flops(G)	BCDD		CDD		LEVIR-CD	
			F1	IoU	F1	IoU	F1	IoU
DWConv	19.61	23.54	92.51	86.06	97.60	95.31	90.74	83.05
Conv	28.77	25.12	92.86	86.67	97.68	95.47	90.71	83.00

6.7%, but the performance increase was very small, so we adopted the DWConv scheme.

V. CONCLUSION

In this article, we propose an end-to-end model-HMCNet for RS. We changed the encoder part of the model to a Siamese structure to facilitate the processing of bitemporal images and introduced MLP into the CNN-based model to form an MLP-CNN hybrid model. HMCNet has both the local feature extraction ability of CNN and the long-range dependence modeling ability of MLP, which can effectively overcome the pseudochange interference in the CD task. The proposed cross-axis attention MLP block induces the local window attention of features through the cross-axis shift window, and the windows are deliberately designed to partially overlap, so the global attention to the features can be formed through the interaction between the information flows in the direction of the two cross-axis, which effectively improves the performance of the MLP block. Our experiments on three public datasets show that HMCNet can outperform many recent CD models with smaller parameters and Flops, and almost outperform these end-to-end models in inference speed. In the process of research, we also found the problem of using more CDD datasets in the CD field and proved that HMCNet has good generalization ability with fewer train data. However, our model still has the problem of slow convergence, so our next work will focus on how to improve the convergence speed of the model to further improve the application ability of the CD model in real-time scenarios.

REFERENCES

[1] D. Amitrano, R. Guida, and P. Iervolino, "Semantic unsupervised change detection of natural land cover with multitemporal object-based analysis on SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5494–5514, Jul. 2021.

[2] C. He, A. Wei, P. Shi, Q. Zhang, and Y. Zhao, "Detecting land-use/land-cover change in rural–urban fringe areas using extended change-vector analysis," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 13, no. 4, pp. 572–585, Aug. 2011.

[3] S. H. Khan, X. He, F. Porikli, and M. Bennamoun, "Forest change detection in incomplete satellite images with deep neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5407–5423, Sep. 2017.

[4] D. A. de Alwis Pitts and E. So, "Enhanced change detection index for disaster response, recovery assessment and monitoring of accessibility and open spaces (camp sites)," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 57, pp. 49–60, May 2017.

[5] P. Washaya, T. Balz, and B. Mohamadi, "Coherence change-detection with Sentinel-1 for natural and anthropogenic disaster monitoring in urban areas," *Remote Sens.*, vol. 10, no. 7, p. 1026, Jun. 2018.

[6] W. Ma, L. Jiao, M. Gong, and C. Li, "Image change detection based on an improved rough fuzzy c-means clustering algorithm," *Int. J. Mach. Learn. Cybern.*, vol. 5, no. 3, pp. 369–377, Jun. 2014.

[7] J. Chen, X. Chen, X. Cui, and J. Chen, "Change vector analysis in posterior probability space: A new method for land cover change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 2, pp. 317–321, Mar. 2011.

[8] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[9] M. Gong, T. Zhan, P. Zhang, and Q. Miao, "Superpixel-based difference representation learning for change detection in multispectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2658–2673, May 2017.

[10] C. Zhang, G. Li, and W. Cui, "High-resolution remote sensing image change detection by statistical-object-based method," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 7, pp. 2440–2447, Jul. 2018.

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[11] Y. Zhang, D. Peng, and X. Huang, "Object-based change detection for VHR images based on multiscale uncertainty analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 1, pp. 13–17, Jan. 2018.
[12] J. Im, J. R. Jensen, and J. A. Tullis, "Object-based change detection using correlation image analysis and image segmentation," *Int. J. Remote Sens.*, vol. 29, no. 2, pp. 399–423, Jan. 2008.
[13] Z. Li, W. Shi, P. Lu, L. Yan, Q. Wang, and Z. Miao, "Landslide mapping from aerial photographs using change detection-based Markov random field," *Remote Sens. Environ.*, vol. 187, pp. 76–90, Dec. 2016.
[14] P. Lu, Y. Qin, Z. Li, A. C. Mondini, and N. Casagli, "Landslide mapping from multi-sensor data through improved change detection-based Markov random field," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111235.
[15] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.
[16] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 2, pp. 84–90, Jun. 2012.
[17] A. Vaswani et al., "Attention is all you need," 2017, *arXiv:1706.03762*.
[18] F. Rosenblatt, "The perceptron: A probabilistic model for information storage and organization in the brain," *Psychol. Rev.*, vol. 65, no. 6, pp. 386–408, 1958.
[19] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[21] W. Wang, Q. Lai, H. Fu, J. Shen, H. Ling, and R. Yang, "Salient object detection in the deep learning era: An in-depth survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 6, pp. 3239–3259, Jun. 2022.
[22] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 833–851.
[23] N. Mahmoudi, S. M. Ahadi, and M. Rahmati, "Multi-target tracking using CNN-based features: CNNMTT," *Multimedia Tools Appl.*, vol. 78, no. 6, pp. 7077–7096, Mar. 2019.
[24] S. Cheng, L. Wang, A. Du, and Y. Li, "Bidirectional focused semantic alignment attention network for cross-modal retrieval," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 4340–4344.
[25] B. Liu, J. Tang, H. Huang, and X.-Y. Lu, "Deep learning methods for super-resolution reconstruction of turbulent flows," *Phys. Fluids*, vol. 32, no. 2, Feb. 2020, Art. no. 025105.
[26] R. C. Daudt, B. L. Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.
[27] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[28] Z. Zhou et al., "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. 4th Int. Workshop, 8th Int. Workshop (ML-CDS)*, Granada, Spain, vol. 11045, 2018, pp. 3–11.
[29] D. Lu, S. Cheng, L. Wang, and S. Song, "Multi-scale feature progressive fusion network for remote sensing image change detection," *Sci. Rep.*, vol. 12, no. 1, p. 11968, Dec. 2022.
[30] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, pp. 1–64, Jun. 2020.
[31] I. Lauriola, A. Lavelli, and F. Aiolli, "An introduction to deep learning in natural language processing: Models, techniques, and tools," *Neurocomputing*, vol. 470, pp. 443–456, Jan. 2022.
[32] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020: *arXiv:2010.11929*.
[33] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.
[34] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.
[35] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," 2022, *arXiv:2201.01293*.
[36] J. Liu, X. Huang, G. Song, H. Li, and Y. Liu, "UniNet: Unified architecture search with convolution, transformer, and MLP," 2022, *arXiv:2207.05420*.
[37] X.-J. Tang, X. Liu, P.-F. Yan, B.-X. Li, H.-Y. Qi, and F. Huang, "An MLP network based on residual learning for Rice hyperspectral data classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[38] P. W. Battaglia et al., "Relational inductive biases, deep learning, and graph networks," 2018, *arXiv:1806.01261*.
[39] I. Tolstikhin et al., "MLP-Mixer: An all-MLP architecture for vision," 2021 *arXiv:2105.01601*.
[40] S. Chen, E. Xie, C. Ge, R. Chen, D. Liang, and P. Luo, "CycleMLP: A MLP-like architecture for dense prediction," 2021, *arXiv:2107.10224*.
[41] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," 2021, *arXiv:2105.15203*.
[42] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[43] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 565–571, May 2018.
[44] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.
[45] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Auto. Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.
[46] K. Lee et al., "Deduplicating training data makes language models better," 2021 *arXiv:2107.06499*.
[47] T. Xiao, M. Singh, E. Mintun, T. Darrell, P. Dollár, and R. Girshick, "Early convolutions help transformers see better," 2021, *arXiv:2106.14881*.



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